

A264 – Structural Asymmetries: Chirality, Gravitation, and Magnetic Monopoles

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.1 Purpose

[STIPULATION] This document treats three structural “ugliness” of the Standard Model and their T0 explanation: (1) chirality of the weak interaction, (2) gravitation as a separate tensor term, (3) absence of magnetic monopoles. In all three cases FFGFT provides a geometric explanation from the energy field $E_{\text{field}}(x, t)$ and the parameter ξ . Methodological restriction: parts of the derivation are still in development; what counts as proof and what as sketch is explicitly marked.

.2 Chirality: Left-Handedness of the Weak Interaction

[B] In the Standard Model the exclusive left-handedness of the weak interaction is an input. In FFGFT it has a geometric basis that follows from half-integrality on the torus.

Half-integrality and inside/outside asymmetry. Spin- $\frac{1}{2}$ follows from the winding number $w = n_\phi/n_\theta = 1$ (winding (1,1) on T^4 , 720° symmetry). The decisive consequence: the torus has an *inside and outside* with different effective couplings. The electric field on the outside points topologically fixed inward (electron) or outward (positron) – the direction is prescribed by the winding geometry, not freely choosable.

[B] After the projection $T^4 \rightarrow \mathbb{R}^3 \times \mathbb{R}_t$ (A090), inner and outer coupling appear as different chiral states. The projection is not symmetric: the compact topology selects one coupling direction. Left-handed coupling is the outer perspective, right-handed the inner perspective – and the weak interaction couples exclusively to the projected outer state.

[S] What has not yet been carried out. The qualitative mechanism is clear: half-integrality $\Rightarrow$ inside/outside asymmetry $\Rightarrow$ chiral projection. What is missing: an explicit computation showing that $g_R = 0$ (not $g_R = -g_L$) follows from this projection. That is the open quantitative edge – not the basic idea.

.3 Gravitation as Field Gradient

[S] In the SM gravitation is a separate tensor field ($G_{\mu\nu}$). In FFGFT gravitation is the gradient of the same energy field that also describes particle masses. The nonlinear field equation:

$$\square E_{\text{field}} + \xi E_{\text{field}}^3 = 0.$$

[K] Equivalence in the weak field. For $E_{\text{field}} = E_0(1+h)$ with small perturbation h , the field equation linearises to a form that is equivalent to the linearised Einstein equation. The full covariant tensor formulation $g_{\mu\nu}(E_{\text{field}})$ with Riemann and Ricci tensors is worked out in the legacy corpus. This is a **[K]** result: equivalence in the weak field is shown; completeness of the equivalence in the strong field is not at proof level.

.4 Magnetic Monopoles

[S] Magnetic monopoles are absent in the SM; their absence is unexplained. In FFGFT they are topological excitations of the energy field and satisfy the Dirac quantisation condition $q_e q_m = 2\pi n \hbar$. Their rarity is a consequence of the high energy threshold $\sim E_P/\xi \approx 7.5 \times 10^3 E_P$ – far beyond any currently achievable energy.

[S] Status. The topological argument is plausible; a direct proof that the T0 energy-field monopole exactly satisfies (and not merely reproduces) the Dirac quantisation condition is not fully worked out.

.5 Objection-Refutation Table

[K] Standard objections and their status in FFGFT:

Objection	Status	Comment
Dimensional inconsistency	Refuted	Normalisation with E_P , L_0 shown
No GR equivalence	Partial	Weak field rigorous; strong field sketch
Missing nonlinearity	Refuted	E^3 term present, Lagrangian shown
No covariance	Refuted	Tensor $g_{\mu\nu}$ constructed
Monopole problem	Sketch	Topological interpretation, threshold plausible
No renormalisability	Refuted	$[E^4] = 4$, renormalisable

.6 Position in the A-Series

[B] The chirality question is worked out in A095: the inside/outside topology of the torus and the chiral projection connect half-integrality (A263) with the left-handedness of the weak coupling geometrically.

[B] The gravitation-as-field-gradient thesis is related to A142 (time-field Lagrange): there gravitation is derived from the time field; here from the energy field. Both approaches need to be checked for consistency – this has not yet been done.

.7 Verification Script

- Chirality phase difference for left/right-handers; weak-field linearisation; monopole threshold from E_P/ξ : `python/A_Serie_Skripte/a264_asymmetrie.py`

.8 Open Questions

Open quantitative edge: $g_R = 0$ **from the projection.** The mechanism half-integrality $\Rightarrow$ inside/outside asymmetry $\Rightarrow$ chiral projection is geometrically clear. What is still missing: the explicit computation showing that the weak coupling of the projected inner state is exactly zero – and not merely smaller than the outer coupling. That is a quantitative task, not a conceptual one.

Gravitation in the strong field is not at proof level. The weak-field equivalence is shown; the full nonlinear equivalence is missing.

The connection to the projection chain (A090) is not worked out. Whether the weak interaction follows from the decompactification is open.

.9 Supersedes

This document subsumes: Doc. 143 (chirality from energy-field rotation, gravitation as field gradient, magnetic monopoles as topological excitations, field equation $\square E + \xi E^3 = 0$, co-variant tensor), Doc. 144 (objection-refutation table, renormalisability, SM as effective low-energy theory). *Not taken over:* excluded applications from Doc. 143/144 (Λ_{cosmo} value, tension parameter).

.10 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.